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SmartSplit

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Project Abstract

This project proposes the development of a Budget Splitter a web–based application designed to help roommates and shared households manage expenses efficiently. The system will allow users to create households, log shared expenses, split costs using multiple methods, and track balances between members. Unlike existing solutions, this platform will focus on student housing scenarios, offering rent templates, payment reminders, and payment logging. The application aims to reduce financial disputes, improve transparency, and simplify shared financial management for college students.

Conceptual Design (Skip this for initial submission)

See attached for diagrams. Implementation will be with Node.js and Express for the backend and Flutter for the frontend.

Background

In college, a very common issue between roommates is finances which often becomes difficult to handle as it is over complicated by a lack of communication and financial management. SmartSplit aims to reduce these struggles by providing a simple and easy way to manage household budgeting involving rent, utilities, groceries, and shared subscriptions where roommates can choose to split equally, by percentage, or custom amounts. SmartSplit will also include balance tracking, displaying which roommate owes who, net balances, and settlement suggestions with the addition of payment logging of Venmo, Zelle, PayPal, etc transactions. This tool will also implement rent due alerts, bill deadlines, and payment nudges.

Expense splitting software already exists in products such as Splitwise that allow users to divide shared costs. While these platforms provide similiar features, they often include additional financial features or multiple scenario applications that complicate the interface, making it cluttered and overwhelming for users who only need simple roommate expense management. SmartSplit will focus specifically on shared household budgeting rather than full personal finance tracking. The goal is to create a more lightweight and student-friendly tool that prioritizes clarity and ease of use. I aim to accomplish this by having straight-forward features that are solely focused on assisting college students navigate finances and possess no other applications. To keep students who are often struggling to keep up with finances, Smartsplit can predict future upcoming household costs based on previous spending patterns and highlight overspending or increasing costs. Additionally, because this is catered to students, Smartsplit aims to incorporate lease logistics such as lease-aware rent distribution. To address the sticky, awkward situations between roommates, SmartSplit also has reminders for unpaid balances between one another.

Open-source resources such as UI frameworks and database libraries may be used to accelerate development, but the core expense splitting logic, balance calculations, and household management system will be implemented independently. This ensures the project demonstrates original software design while still leveraging existing development tools where appropriate.

Proof of Concept - https://github.com/IsaacCheng9/fairsplit.git

For the proof of concept, I selected an existing open-source project called FairSplit, which is a bill-splitting application using Node.js and React. The repository demonstrates core shared-expense functionality and mainly focuses on a smart settlement system that minimizes the number of

transactions required to balance group debts using a greedy algorithm. This repository serves as a strong proof of concept because it demonstrates that shared expense tracking, balance calculation, and smart debt settlement logic can function together in a working system.

Required Resources

For software resources, I'll be using Node.js for the backend and React or Flutter for the frontend, along with a database such as MySQL to store user and expense data. Development will also require Git/GitHub for version control.

Hardware requirements are minimal since everything can be developed on a standard laptop or desktop capable of running a code editor and local server. No specialized equipment is needed, and the project fits within typical Computer Science lab resources.